POWER BI CAPSTONE · 2025

Hotel Performance Analytics Dashboard

Data-Driven Insights for Hotel Revenue, Bookings, Guest Experience, and Operational Performance

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Project Overview

This project analyzes hotel booking and operational data using Power BI to surface actionable insights across revenue, demand, and guest experience.

INSIGHTS COVERED

- Hotel revenue performance
- Room category demand
- Booking channels
- Cancellations and booking status
- Booking volume
- Guest ratings
- Hotel locations

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Key Performance Indicators

AVERAGE DAILY RATE

$193.83

ADR

TOTAL BOOKING REVENUE

$6.56M

Revenue

HIGHEST REVENUE

$2.56M

Top Segment

TOTAL BOOKINGS

7K

Reservations

TOTAL NIGHTS

34K

Room Nights

STATUS COUNT

7K

All Records

CANCELLED BOOKINGS

1K

≈ 17%

SUMMARY

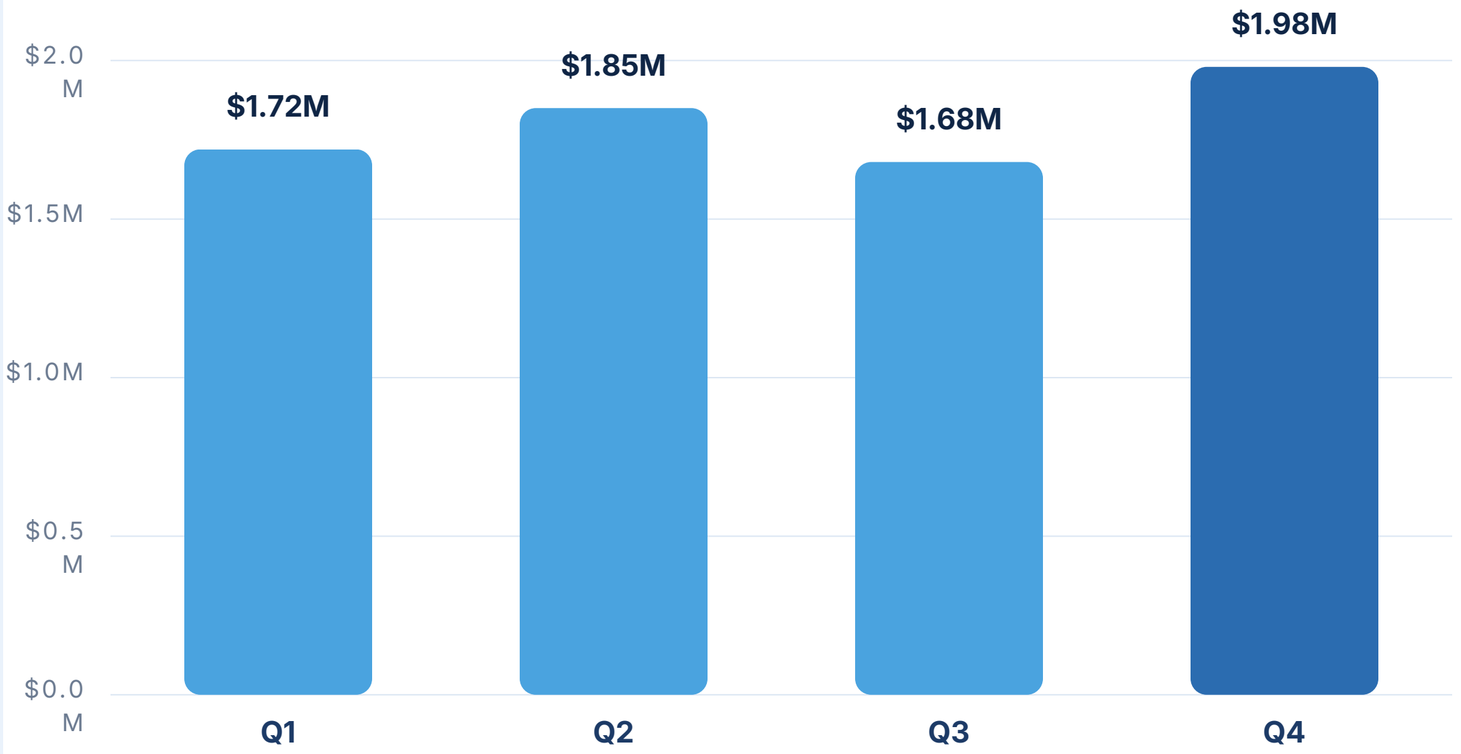
The hotel portfolio generated strong revenue and maintained a high volume of bookings, while cancellations represent an important area for improvement.

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Revenue Performance

Quarterly Booking Revenue

FY total ≈ $6.56M



KEY INSIGHT

Total booking revenue by quarter remains stable throughout the year, with quarterly values close to $1.5M–$2M — indicating consistent demand and stable hotel performance.

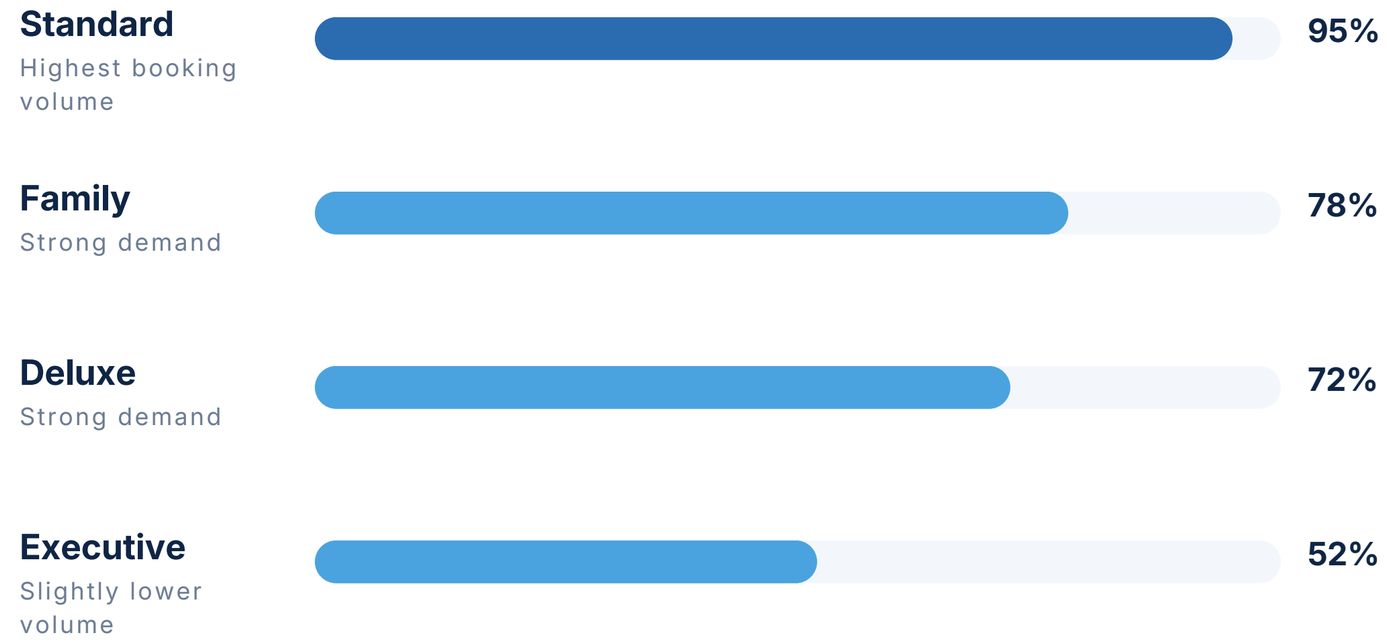
$1.81M

AVG QUARTERLY REVENUE

04

Room Category Analysis

Booking Volume by Room Category



KEY INSIGHT

Standard rooms are the primary driver of booking volume and should remain a major focus for capacity planning and marketing.

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Hotel Location Performance

Cairo

TOTAL NIGHTS
6.4K

REVENUE
$1.24M

Alexandria

TOTAL NIGHTS
5.8K

REVENUE
$1.05M

Sharm El Sheikh

TOTAL NIGHTS
6.9K

REVENUE
$1.42M

Hurghada

TOTAL NIGHTS
5.2K

REVENUE
$0.98M

Luxor

TOTAL NIGHTS
4.8K

REVENUE
$0.92M

Aswan

TOTAL NIGHTS
4.9K

REVENUE
$0.95M

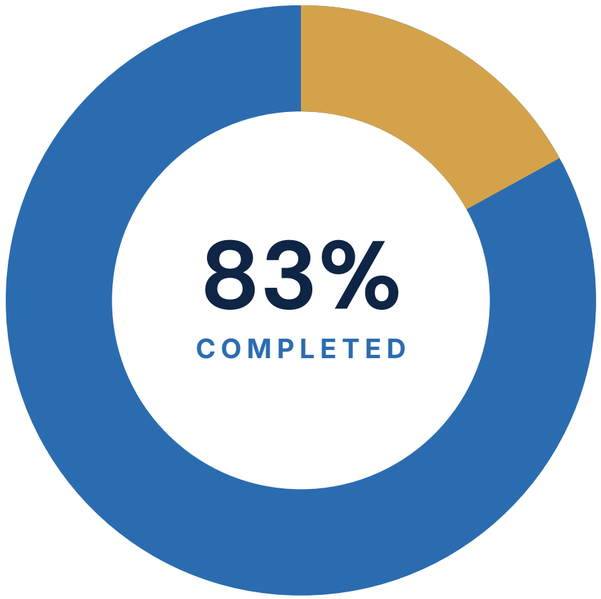
KEY INSIGHT

The hotel network has diversified demand across destinations, reducing dependence on any single location.

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Booking Status and Cancellations

Reservation Outcomes



Completed bookings	83%
No-show / Cancelled	17%

KEY INSIGHT

Most reservations are successfully completed, but reducing cancellations and no-shows could meaningfully improve revenue and occupancy.

5.8K
COMPLETED BOOKINGS

1.2K
CANCELLED / NO-SHOW

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Booking Channel Analysis

DIRECT WEBSITE

25%

Owned digital channel

PHONE

25%

Voice reservations

OTA

25%

Online travel agencies

CORPORATE

25%

Business accounts

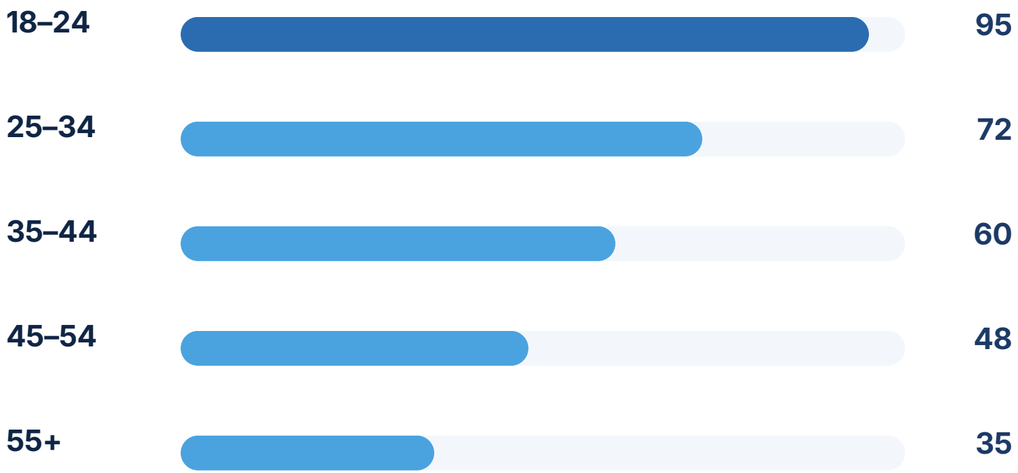
KEY INSIGHT

Revenue is evenly distributed across booking channels, creating a balanced and resilient sales mix.

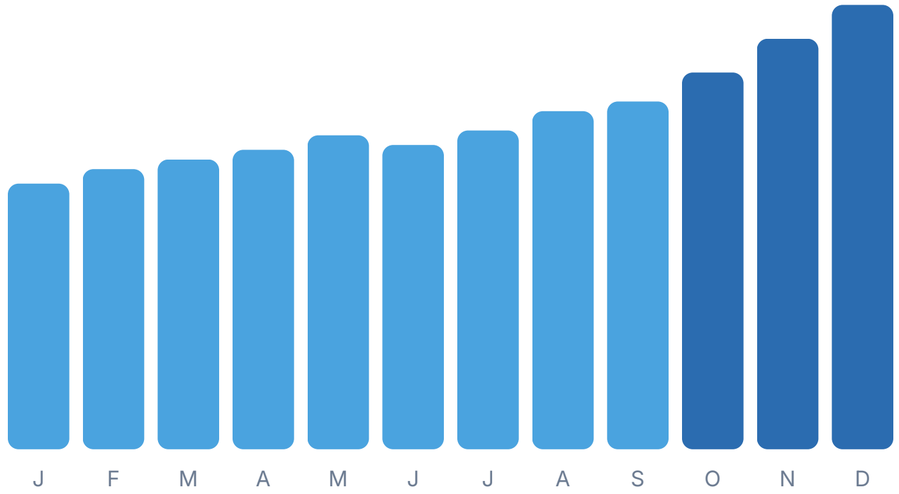
08

Guest Demographics and Activity

Guests by Age Group



Monthly Guest Activity



KEY INSIGHT

Young travelers (18–24) represent an important customer segment and may respond well to digital promotions and mobile-first booking experiences.

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Guest Satisfaction

AVERAGE GUEST RATING

3.80

OUT OF 5.00

Guest satisfaction is generally stable, but improving service quality could help increase ratings above 4.0.

Rating by Location



Business Recommendations

1 Reduce Cancellations

Deploy reminder messages and flexible booking policies to lower no-show rates.

2 Boost Room Marketing

Increase demand generation for high-performing Standard and Family rooms.

3 Elevate Guest Experience

Target service improvements to lift the average rating above 4.0.

4 Reach Younger Travelers

Run targeted digital campaigns and mobile-first offers for 18–24 guests.

5 Balance Channel Mix

Maintain balanced investment across Direct, Phone, OTA, and Corporate.

6 Replicate Top Locations

Study high-performing hotels and apply their practices across the network.

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Conclusion

Key Takeaways

1. Strong revenue generation
2. Consistent booking activity
3. Balanced performance across locations and channels
4. High completion rates
5. Opportunities to reduce cancellations
6. Opportunities to improve guest satisfaction

FINAL STATEMENT

Data-driven decision-making can help hotels improve revenue, occupancy, customer satisfaction, and operational efficiency.

Thank You

Questions & Discussion